1.2 Lengths and Dot Products

The first section backed off from multiplying vectors. Now we go forward to define the *"dot product"* of v and w. This multiplication involves the separate products v_1w_1 and v_2w_2, but it doesn't stop there. Those two numbers are added to produce the single number $v \cdot w$. *This is the geometry section (lengths and angles).*

DEFINITION The ***dot product*** or ***inner product*** of $v = (v_1, v_2)$ and $w = (w_1, w_2)$ is the number $v \cdot w$:

$$v \cdot w = v_1w_1 + v_2w_2. \tag{1}$$

Example 1 The vectors $v = (4, 2)$ and $w = (-1, 2)$ have a *zero* dot product:

Dot product is zero
Perpendicular vectors

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = -4 + 4 = 0.$$

In mathematics, zero is always a special number. For dot products, it means that *these two vectors are perpendicular.* The angle between them is 90°. When we drew them in Figure 1.1, we saw a rectangle (not just any parallelogram). The clearest example of perpendicular vectors is $i = (1, 0)$ along the x axis and $j = (0, 1)$ up the y axis. Again the dot product is $i \cdot j = 0 + 0 = 0$. Those vectors i and j form a right angle.

The dot product of $v = (1, 2)$ and $w = (3, 1)$ is 5. Soon $v \cdot w$ will reveal the angle between v and w (not 90°). Please check that $w \cdot v$ is also 5.

The dot product $w \cdot v$ ***equals*** $v \cdot w$. The order of v and w makes no difference.

Example 2 Put a weight of 4 at the point $x = -1$ (left of zero) and a weight of 2 at the point $x = 2$ (right of zero). The x axis will balance on the center point (like a see-saw). The weights balance because the dot product is $(4)(-1) + (2)(2) = 0$.

This example is typical of engineering and science. The vector of weights is $(w_1, w_2) = (4, 2)$. The vector of distances from the center is $(v_1, v_2) = (-1, 2)$. The weights times the distances, w_1v_1 and w_2v_2, give the "moments". The equation for the see-saw to balance is $w_1v_1 + w_2v_2 = 0$.

Example 3 Dot products enter in economics and business. We have three goods to buy and sell. Their prices are (p_1, p_2, p_3) for each unit—this is the "price vector" p. The quantities we buy or sell are (q_1, q_2, q_3)—positive when we sell, negative when we buy. *Selling* q_1 *units at the price* p_1 *brings in* q_1p_1. The total income (quantities q times prices p) is ***the dot product*** $q \cdot p$ ***in three dimensions***:

$$\textit{Income} = (q_1, q_2, q_3) \cdot (p_1, p_2, p_3) = q_1p_1 + q_2p_2 + q_3p_3 = \textit{dot product.}$$

A zero dot product means that "the books balance". Total sales equal total purchases if $q \cdot p = 0$. Then p is perpendicular to q (in three-dimensional space). A supermarket with thousands of goods goes quickly into high dimensions.

Small note: Spreadsheets have become essential in management. They compute linear combinations and dot products. What you see on the screen is a matrix.

Main point To compute $v \cdot w$, multiply each v_i times w_i. Then add $\Sigma v_i w_i$.

Lengths and Unit Vectors

An important case is the dot product of a vector *with itself.* In this case v equals w. When the vector is $v = (1, 2, 3)$, the dot product with itself is $v \cdot v = \|v\|^2 = 14$:

Dot product $v \cdot v$
Length squared

$$\|v\|^2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 + 4 + 9 = \mathbf{14}.$$

Instead of a 90° angle between vectors we have 0°. The answer is not zero because v is not perpendicular to itself. The dot product $v \cdot v$ gives the *length of v squared.*

DEFINITION The *length* $\|v\|$ of a vector v is the square root of $v \cdot v$:

Length = norm(v) $\qquad$ **length** $= \|v\| = \sqrt{v \cdot v}$.

In two dimensions the length is $\sqrt{v_1^2 + v_2^2}$. In three dimensions it is $\sqrt{v_1^2 + v_2^2 + v_3^2}$. By the calculation above, the length of $v = (1, 2, 3)$ is $\|v\| = \sqrt{14}$.

Here $\|v\| = \sqrt{v \cdot v}$ is just the ordinary length of the arrow that represents the vector. In two dimensions, the arrow is in a plane. If the components are 1 and 2, the arrow is the third side of a right triangle (Figure 1.6). The Pythagoras formula $a^2 + b^2 = c^2$, which connects the three sides, is $1^2 + 2^2 = \|v\|^2$.

For the length of $v = (1, 2, 3)$, we used the right triangle formula twice. The vector $(1, 2, 0)$ in the base has length $\sqrt{5}$. This base vector is perpendicular to $(0, 0, 3)$ that goes straight up. So the diagonal of the box has length $\|v\| = \sqrt{5 + 9} = \sqrt{14}$.

The length of a four-dimensional vector would be $\sqrt{v_1^2 + v_2^2 + v_3^2 + v_4^2}$. Thus the vector $(1, 1, 1, 1)$ has length $\sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2$. This is the diagonal through a unit cube in four-dimensional space. The diagonal in n dimensions has length $\sqrt{n}$.

The word "**unit**" is always indicating that some measurement equals "one". The unit price is the price for one item. A unit cube has sides of length one. A unit circle is a circle with radius one. Now we define the idea of a "unit vector".

DEFINITION *A unit vector u is a vector whose length equals one.* Then $u \cdot u = 1$.

An example in four dimensions is $u = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Then $u \cdot u$ is $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$. We divided $v = (1, 1, 1, 1)$ by its length $\|v\| = 2$ to get this unit vector.

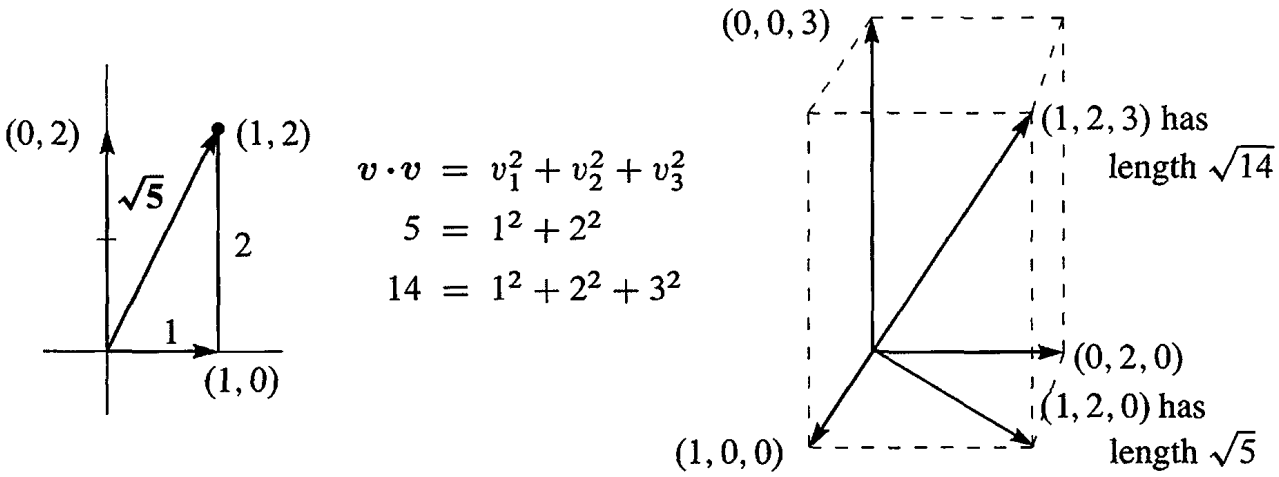


Figure 1.6: The length $\sqrt{v \cdot v}$ of two-dimensional and three-dimensional vectors.

Example 4 The standard unit vectors along the x and y axes are written $\boldsymbol{i}$ and $\boldsymbol{j}$. In the xy plane, the unit vector that makes an angle "theta" with the x axis is $(\cos\theta, \sin\theta)$:

$$\textbf{Unit vectors} \quad \boldsymbol{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \boldsymbol{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{u} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}.$$

When $\theta = 0$, the horizontal vector $\boldsymbol{u}$ is $\boldsymbol{i}$. When $\theta = 90°$ (or $\frac{\pi}{2}$ radians), the vertical vector is $\boldsymbol{j}$. At any angle, the components $\cos\theta$ and $\sin\theta$ produce $\boldsymbol{u} \cdot \boldsymbol{u} = 1$ because $\cos^2\theta + \sin^2\theta = 1$. These vectors reach out to the unit circle in Figure 1.7. Thus $\cos\theta$ and $\sin\theta$ are simply the coordinates of that point at angle θ on the unit circle.

Since $(2, 2, 1)$ has length 3, the vector $(\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$ has length 1. Check that $\boldsymbol{u} \cdot \boldsymbol{u} = \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1$. For a unit vector, **divide any nonzero v by its length $\|v\|$.**

Unit vector $\boldsymbol{u} = v/\|v\|$ **is a unit vector in the same direction as v.**

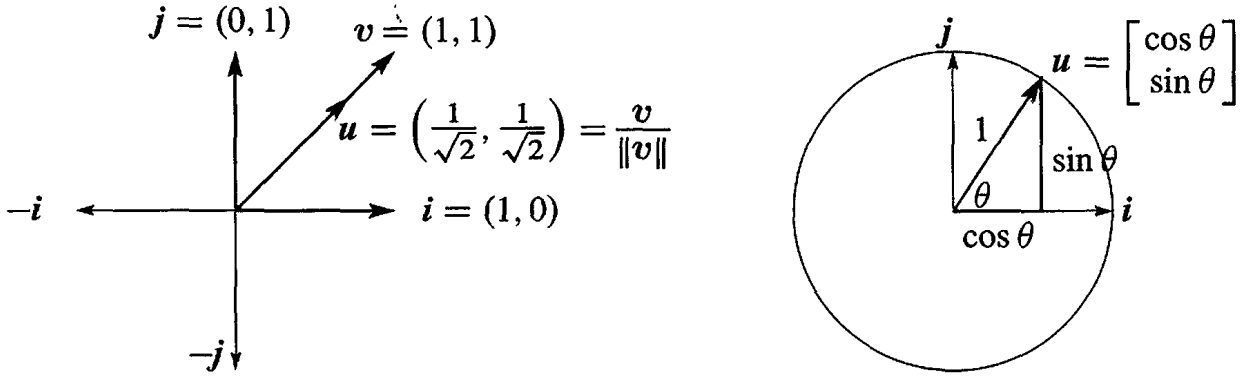


Figure 1.7: The coordinate vectors $\boldsymbol{i}$ and $\boldsymbol{j}$. The unit vector $\boldsymbol{u}$ at angle 45° (left) divides $v = (1, 1)$ by its length $\|v\| = \sqrt{2}$. The unit vector $\boldsymbol{u} = (\cos\theta, \sin\theta)$ is at angle θ.

The Angle Between Two Vectors

We stated that perpendicular vectors have $v \cdot w = 0$. The dot product is zero when the angle is 90°. To explain this, we have to connect angles to dot products. Then we show how $v \cdot w$ finds the angle between any two nonzero vectors v and w.

Right angles *The dot product is $v \cdot w = 0$ when v is perpendicular to w.*

Proof When v and w are perpendicular, they form two sides of a right triangle. The third side is $v - w$ (the hypotenuse going across in Figure 1.8). The *Pythagoras Law* for the sides of a right triangle is $a^2 + b^2 = c^2$:

$$\textbf{Perpendicular vectors} \quad \|v\|^2 + \|w\|^2 = \|v - w\|^2 \tag{2}$$

Writing out the formulas for those lengths in two dimensions, this equation is

$$\textbf{Pythagoras} \quad (v_1^2 + v_2^2) + (w_1^2 + w_2^2) = (v_1 - w_1)^2 + (v_2 - w_2)^2. \tag{3}$$

The right side begins with $v_1^2 - 2v_1w_1 + w_1^2$. Then v_1^2 and w_1^2 are on both sides of the equation and they cancel, leaving $-2v_1w_1$. Also v_2^2 and w_2^2 cancel, leaving $-2v_2w_2$. (In three dimensions there would be $-2v_3w_3$.) Now divide by -2:

$$0 = -2v_1w_1 - 2v_2w_2 \quad \text{which leads to} \quad \boldsymbol{v_1w_1 + v_2w_2 = 0}. \tag{4}$$

Conclusion Right angles produce $v \cdot w = 0$. The dot product is zero when the angle is $\theta = 90°$. Then $\cos\theta = 0$. The zero vector $v = \mathbf{0}$ is perpendicular to every vector w because $\mathbf{0} \cdot w$ is always zero.

Now suppose $v \cdot w$ is **not zero**. It may be positive, it may be negative. The sign of $v \cdot w$ immediately tells whether we are below or above a right angle. The angle is less than 90° when $v \cdot w$ is positive. The angle is above 90° when $v \cdot w$ is negative. The right side of Figure 1.8 shows a typical vector $v = (3, 1)$. The angle with $w = (1, 3)$ is less than 90° because $v \cdot w = 6$ is positive.

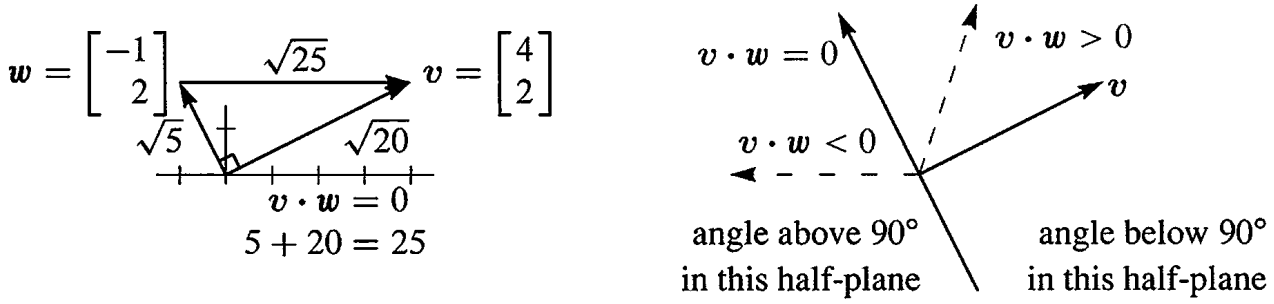


Figure 1.8: Perpendicular vectors have $v \cdot w = 0$. Then $\|v\|^2 + \|w\|^2 = \|v - w\|^2$.

The borderline is where vectors are perpendicular to v. On that dividing line between plus and minus, $(1, -3)$ is perpendicular to $(3, 1)$. The dot product is zero.

The dot product reveals the exact angle θ. This is not necessary for linear algebra—you could stop here! Once we have matrices, we won't come back to θ. But while we are on the subject of angles, this is the place for the formula.

Start with **unit vectors u and U**. The sign of $u \cdot U$ tells whether $\theta < 90°$ or $\theta > 90°$. Because the vectors have length 1, we learn more than that. ***The dot product $u \cdot U$ is the cosine of θ.*** This is true in any number of dimensions.

Unit vectors u and U at angle θ have $u \cdot U = \cos\theta$. Certainly $|u \cdot U| \leq 1$.

Remember that $\cos\theta$ is never greater than 1. It is never less than -1. ***The dot product of unit vectors is between -1 and 1.***

Figure 1.9 shows this clearly when the vectors are $u = (\cos\theta, \sin\theta)$ and $i = (1, 0)$. The dot product is $u \cdot i = \cos\theta$. That is the cosine of the angle between them.

After rotation through any angle α, these are still unit vectors. The vector $i = (1, 0)$ rotates to $(\cos\alpha, \sin\alpha)$. The vector u rotates to $(\cos\beta, \sin\beta)$ with $\beta = \alpha + \theta$. Their dot product is $\cos\alpha\cos\beta + \sin\alpha\sin\beta$. From trigonometry this is the same as $\cos(\beta - \alpha)$. But $\beta - \alpha$ is the angle θ, so the dot product is $\cos\theta$.

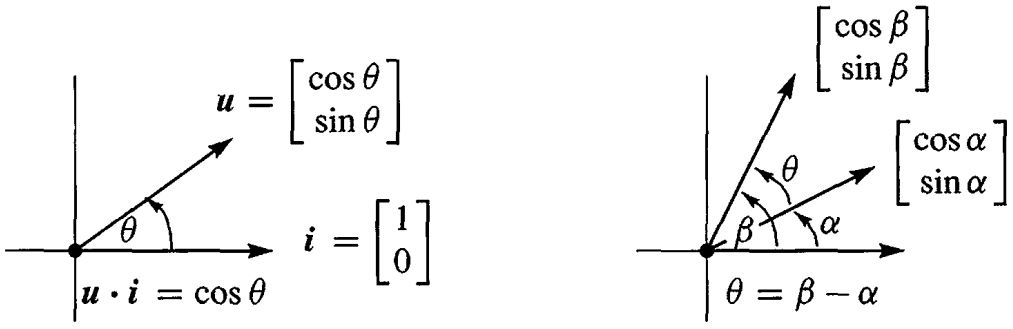


Figure 1.9: The dot product of unit vectors is the cosine of the angle θ.

Problem 24 proves $|u \cdot U| \leq 1$ directly, without mentioning angles. The inequality and the cosine formula $u \cdot U = \cos\theta$ are always true for unit vectors.

What if v and w are not unit vectors? Divide by their lengths to get $u = v/\|v\|$ and $U = w/\|w\|$. Then the dot product of those unit vectors u and U gives $\cos\theta$.

COSINE FORMULA If v and w are nonzero vectors then $\dfrac{v \cdot w}{\|v\|\,\|w\|} = \cos\theta$.

Whatever the angle, this dot product of $v/\|v\|$ with $w/\|w\|$ never exceeds one. That is the ***"Schwarz inequality"*** $|v \cdot w| \leq \|v\| \, \|w\|$ for dot products—or more correctly the Cauchy-Schwarz-Buniakowsky inequality. It was found in France and Germany and Russia (and maybe elsewhere—it is the most important inequality in mathematics).

Since $|\cos\theta|$ never exceeds 1, the cosine formula gives two great inequalities:

SCHWARZ INEQUALITY $\quad |v \cdot w| \leq \|v\| \, \|w\|$

TRIANGLE INEQUALITY $\quad \|v + w\| \leq \|v\| + \|w\|$

Example 5 Find $\cos\theta$ for $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and check both inequalities.

Solution The dot product is $v \cdot w = 4$. Both v and w have length $\sqrt{5}$. The cosine is $4/5$.

$$\cos\theta = \frac{v \cdot w}{\|v\| \, \|w\|} = \frac{4}{\sqrt{5}\sqrt{5}} = \frac{4}{5}.$$

The angle is below 90° because $v \cdot w = 4$ is positive. By the Schwarz inequality, $v \cdot w = 4$ is less than $\|v\| \, \|w\| = 5$. Side $3 = \|v + w\|$ is less than side 1 + side 2, by the triangle inequality. For $v + w = (3, 3)$ that says $\sqrt{18} < \sqrt{5} + \sqrt{5}$. Square this to get $18 < 20$.

Example 6 The dot product of $v = (a, b)$ and $w = (b, a)$ is $2ab$. Both lengths are $\sqrt{a^2 + b^2}$. The Schwarz inequality in this case says that $2ab \leq a^2 + b^2$.

This is more famous if we write $x = a^2$ and $y = b^2$. The "geometric mean" $\sqrt{xy}$ is not larger than the "arithmetic mean" = average $\frac{1}{2}(x + y)$.

$$\begin{matrix} \textbf{Geometric} \\ \textbf{mean} \end{matrix} \leq \begin{matrix} \textbf{Arithmetic} \\ \textbf{mean} \end{matrix} \qquad ab \leq \frac{a^2 + b^2}{2} \quad \text{becomes} \quad \sqrt{xy} \leq \frac{x + y}{2}.$$

Example 5 had $a = 2$ and $b = 1$. So $x = 4$ and $y = 1$. The geometric mean $\sqrt{xy} = 2$ is below the arithmetic mean $\frac{1}{2}(1 + 4) = 2.5$.

Notes on Computing

Write the components of v as $v(1), \ldots, v(N)$ and similarly for w. In FORTRAN, the sum $v + w$ requires a loop to add components separately. The dot product also uses a loop to add the separate $v(j)w(j)$. Here are VPLUSW and VDOTW:

FORTRAN

```
DO 10 J = 1,N                          DO 10 J = 1,N
10 VPLUSW(J) = v(J) + w(J)             10 VDOTW = VDOTW + V(J) * W(J)
```

MATLAB *and also* PYTHON *work directly with whole vectors, not their components.* No loop is needed. When v and w have been defined, $v + w$ is immediately understood.

Input v and w as rows—the prime $'$ transposes them to columns. $2v + 3w$ uses $*$ for multiplication by 2 and 3. The result will be printed unless the line ends in a semicolon.

MATLAB $v = [2 \quad 3 \quad 4]' \;\; ; \;\; w = [1 \quad 1 \quad 1]' \;\; ; \;\; u = 2 * v + 3 * w$

The dot product $v \cdot w$ is usually seen as ***a row times a column (with no dot)***:

Instead of $\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}3\\4\end{bmatrix}$ we more often see $[1\ 2]\begin{bmatrix}3\\4\end{bmatrix}$ or $v' * w$

The length of v is known to MATLAB as norm (v). We could define it ourselves as sqrt $(v' * v)$, using the square root function—also known. The cosine we have to define ourselves! The angle (in radians) comes from the *arc cosine* (acos) function:

Cosine formula	cosine $= v' * w/(\text{norm}\,(v) * \text{norm}\,(w))$
Angle formula	angle $=$ acos (cosine)

An M-file would create a new function **cosine** (v, w) for future use. The M-files created especially for this book are listed at the end. R and PYTHON are open source software.

■ REVIEW OF THE KEY IDEAS ■

1. The dot product $v \cdot w$ multiplies each component v_i by w_i and adds all $v_i w_i$.
2. The length $\|v\|$ of a vector is the square root of $v \cdot v$.
3. $u = v/\|v\|$ is a ***unit vector.*** Its length is 1.
4. The dot product is $v \cdot w = 0$ when vectors v and w are perpendicular.
5. The cosine of θ (the angle between any nonzero v and w) never exceeds 1:

$$\cos\theta = \frac{v \cdot w}{\|v\|\|w\|} \qquad \textit{Schwarz inequality} \quad |v \cdot w| \le \|v\|\,\|w\|.$$

Problem 21 will produce the ***triangle inequality*** $\|v + w\| \le \|v\| + \|w\|$.

■ WORKED EXAMPLES ■

1.2 A For the vectors $v = (3, 4)$ and $w = (4, 3)$ test the Schwarz inequality on $v \cdot w$ and the triangle inequality on $\|v + w\|$. Find $\cos\theta$ for the angle between v and w. When will we have *equality* $|v \cdot w| = \|v\|\,\|w\|$ and $\|v + w\| = \|v\| + \|w\|$?

Solution The dot product is $v \cdot w = (3)(4) + (4)(3) = 24$. The length of v is $\|v\| = \sqrt{9+16} = 5$ and also $\|w\| = 5$. The sum $v + w = (7,7)$ has length $7\sqrt{2} < 10$.

Schwarz inequality	$\|v \cdot w\| \le \|v\| \, \|w\|$ is $24 < 25$.
Triangle inequality	$\|v + w\| \le \|v\| + \|w\|$ is $7\sqrt{2} < 5 + 5$.
Cosine of angle	$\cos\theta = \frac{24}{25}$ Thin angle from $v = (3,4)$ to $w = (4,3)$

Suppose one vector is a multiple of the other as in $w = cv$. Then the angle is $0°$ or $180°$. In this case $|\cos\theta| = 1$ and $|v \cdot w|$ *equals* $\|v\| \, \|w\|$. If the angle is $0°$, as in $w = 2v$, then $\|v + w\| = \|v\| + \|w\|$. The triangle is completely flat.

1.2 B Find a unit vector u in the direction of $v = (3,4)$. Find a unit vector U that is perpendicular to u. How many possibilities for U?

Solution For a unit vector u, divide v by its length $\|v\| = 5$. For a perpendicular vector V we can choose $(-4,3)$ since the dot product $v \cdot V$ is $(3)(-4) + (4)(3) = 0$. For a *unit* vector U, divide V by its length $\|V\|$:

$$u = \frac{v}{\|v\|} = \left(\frac{3}{5}, \frac{4}{5}\right) \qquad U = \frac{V}{\|V\|} = \left(-\frac{4}{5}, \frac{3}{5}\right) \qquad u \cdot U = 0$$

The only other perpendicular unit vector would be $-U = (\frac{4}{5}, -\frac{3}{5})$.

1.2 C Find a vector $x = (c,d)$ that has dot products $x \cdot r = 1$ and $x \cdot s = 0$ with the given vectors $r = (2,-1)$ and $s = (-1,2)$.

How is this question related to Example **1.1 C**, which solved $cv + dw = b = (1,0)$?

Solution Those two dot products give linear equations for c and d. Then $x = (c,d)$.

$x \cdot r = 1$	$2c - d = 1$	**The same equations as**
$x \cdot s = 0$	$-c + 2d = 0$	**in Worked Example 1.1 C**

The second equation makes x perpendicular to $s = (-1,2)$. So I can see the geometry: Go in the perpendicular direction $(2,1)$. When you reach $x = \frac{1}{3}(2,1)$, the dot product with $r = (2,-1)$ has the required value $x \cdot r = 1$.

Comment on n equations for $x = (x_1, \ldots, x_n)$ in n-dimensional space

Section 1.1 would start with column vectors $v_1, \ldots, v_n$. The goal is to combine them to produce a required vector $x_1 v_1 + \cdots + x_n v_n = b$. This section would start from vectors $r_1, \ldots, r_n$. Now the goal is to find x with the required dot products $x \cdot r_i = b_i$.

Soon the v's will be the columns of a matrix A, and the r's will be the rows of A. Then the (one and only) problem will be to solve $Ax = b$.

Problem Set 1.2

1 Calculate the dot products $u \cdot v$ and $u \cdot w$ and $u \cdot (v + w)$ and $w \cdot v$:

$$u = \begin{bmatrix} -.6 \\ .8 \end{bmatrix} \qquad v = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \qquad w = \begin{bmatrix} 8 \\ 6 \end{bmatrix}.$$

2 Compute the lengths $\|u\|$ and $\|v\|$ and $\|w\|$ of those vectors. Check the Schwarz inequalities $|u \cdot v| \leq \|u\| \, \|v\|$ and $|v \cdot w| \leq \|v\| \, \|w\|$.

3 Find unit vectors in the directions of v and w in Problem 1, and the cosine of the angle θ. Choose vectors a, b, c that make 0°, 90°, and 180° angles with w.

4 For any *unit* vectors v and w, find the dot products (actual numbers) of

(a) v and $-v$ (b) $v + w$ and $v - w$ (c) $v - 2w$ and $v + 2w$

5 Find unit vectors u_1 and u_2 in the directions of $v = (3, 1)$ and $w = (2, 1, 2)$. Find unit vectors U_1 and U_2 that are perpendicular to u_1 and u_2.

6 (a) Describe every vector $w = (w_1, w_2)$ that is perpendicular to $v = (2, -1)$.

(b) The vectors that are perpendicular to $V = (1, 1, 1)$ lie on a _____.

(c) The vectors that are perpendicular to $(1, 1, 1)$ and $(1, 2, 3)$ lie on a _____.

7 Find the angle θ (from its cosine) between these pairs of vectors:

(a) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (b) $v = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

(c) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$ (d) $v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$.

8 True or false (give a reason if true or a counterexample if false):

(a) If u is perpendicular (in three dimensions) to v and w, those vectors v and w are parallel.

(b) If u is perpendicular to v and w, then u is perpendicular to $v + 2w$.

(c) If u and v are perpendicular unit vectors then $\|u - v\| = \sqrt{2}$.

9 The slopes of the arrows from $(0, 0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1. **Suppose the product $v_2 w_2 / v_1 w_1$ of those slopes is -1.** Show that $v \cdot w = 0$ and the vectors are perpendicular.

10 Draw arrows from $(0, 0)$ to the points $v = (1, 2)$ and $w = (-2, 1)$. Multiply their slopes. That answer is a signal that $v \cdot w = 0$ and the arrows are _____.

11 If $v \cdot w$ is negative, what does this say about the angle between v and w? Draw a 3-dimensional vector v (an arrow), and show where to find all w's with $v \cdot w < 0$.

12 With $v = (1, 1)$ and $w = (1, 5)$ choose a number c so that $w - cv$ is perpendicular to v. Then find the formula that gives this number c for any nonzero v and w. (*Note*: cv is the "projection" of w onto v.)

13 Find two vectors v and w that are perpendicular to $(1, 0, 1)$ and to each other.

14 Find nonzero vectors u, v, w that are perpendicular to $(1, 1, 1, 1)$ and to each other.

15 The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x + y) =$ _____. This would come in Example 6 from the Schwarz inequality for $v = (\sqrt{2}, \sqrt{8})$ and $w = (\sqrt{8}, \sqrt{2})$. Find $\cos\theta$ for this v and w.

16 **How long is the vector $v = (1, 1, \ldots, 1)$ in 9 dimensions?** Find a unit vector u in the same direction as v and a unit vector w that is perpendicular to v.

17 What are the cosines of the angles α, β, θ between the vector $(1, 0, -1)$ and the unit vectors i, j, k along the axes? Check the formula $\cos^2\alpha + \cos^2\beta + \cos^2\theta = 1$.

Problems 18–31 lead to the main facts about lengths and angles in triangles.

18 The parallelogram with sides $v = (4, 2)$ and $w = (-1, 2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$ which is for ***right triangles only***:

$$(\text{length of } v)^2 + (\text{length of } w)^2 = (\text{length of } v + w)^2.$$

19 (Rules for dot products) These equations are simple but useful:

(1) $v \cdot w = w \cdot v$ **(2)** $u \cdot (v + w) = u \cdot v + u \cdot w$ **(3)** $(cv) \cdot w = c(v \cdot w)$

Use **(2)** with $u = v + w$ to prove $\|v + w\|^2 = v \cdot v + 2v \cdot w + w \cdot w$.

20 The "Law of Cosines" comes from $(v - w) \cdot (v - w) = v \cdot v - 2v \cdot w + w \cdot w$:

Cosine Law $$\|v - w\|^2 = \|v\|^2 - 2\|v\|\,\|w\|\cos\theta + \|w\|^2.$$

If $\theta < 90°$ show that $\|v\|^2 + \|w\|^2$ is larger than $\|v - w\|^2$ (the third side).

21 The ***triangle inequality*** says: (length of $v + w$) $\leq$ (length of v) + (length of w).

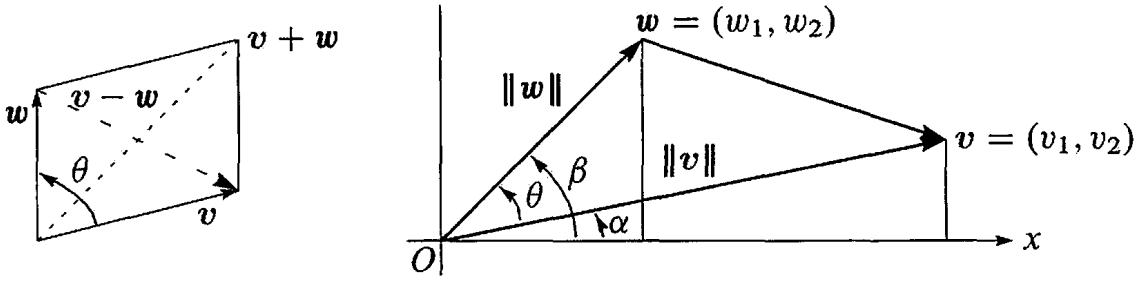
Problem 19 found $\|v + w\|^2 = \|v\|^2 + 2v \cdot w + \|w\|^2$. Use the Schwarz inequality $v \cdot w \leq \|v\|\,\|w\|$ to show that $\|\textbf{side 3}\|$ can not exceed $\|\textbf{side 1}\| + \|\textbf{side 2}\|$:

Triangle inequality $$\|v + w\|^2 \leq (\|v\| + \|w\|)^2 \quad \textbf{or} \quad \|v + w\| \leq \|v\| + \|w\|.$$

22 The Schwarz inequality $|v \cdot w| \leq \|v\|\,\|w\|$ by algebra instead of trigonometry:

(a) Multiply out both sides of $(v_1w_1 + v_2w_2)^2 \leq (v_1^2 + v_2^2)(w_1^2 + w_2^2)$.

(b) Show that the difference between those two sides equals $(v_1w_2 - v_2w_1)^2$. This cannot be negative since it is a square—so the inequality is true.



23 The figure shows that $\cos\alpha = v_1/\|v\|$ and $\sin\alpha = v_2/\|v\|$. Similarly $\cos\beta$ is _____ and $\sin\beta$ is _____. The angle θ is $\beta - \alpha$. Substitute into the trigonometry formula $\cos\beta\cos\alpha + \sin\beta\sin\alpha$ for $\cos(\beta - \alpha)$ to find $\cos\theta = v \cdot w/\|v\|\,\|w\|$.

24 One-line proof of the Schwarz inequality $|u \cdot U| \le 1$ for unit vectors:

$$|u \cdot U| \le |u_1|\,|U_1| + |u_2|\,|U_2| \le \frac{u_1^2 + U_1^2}{2} + \frac{u_2^2 + U_2^2}{2} = \frac{1+1}{2} = 1.$$

Put $(u_1, u_2) = (.6, .8)$ and $(U_1, U_2) = (.8, .6)$ in that whole line and find $\cos\theta$.

25 Why is $|\cos\theta|$ never greater than 1 in the first place?

26 If $v = (1, 2)$ draw all vectors $w = (x, y)$ in the xy plane with $v \cdot w = x + 2y = 5$. Which is the shortest w?

27 (*Recommended*) If $||v|| = 5$ and $||w|| = 3$, what are the smallest and largest values of $||v - w||$? What are the smallest and largest values of $v \cdot w$?

Challenge Problems

28 Can three vectors in the xy plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$? I don't know how many vectors in xyz space can have all negative dot products. (Four of those vectors in the plane would certainly be impossible . . .).

29 Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x, y, z)$ and the vector $w = (z, x, y)$. Challenge question: Explain why $v \cdot w/\|v\|\,\|w\|$ is always $-\frac{1}{2}$.

30 How could you prove $\sqrt[3]{xyz} \le \frac{1}{3}(x+y+z)$ (geometric mean $\le$ arithmetic mean)?

31 Find four perpendicular unit vectors with all components equal to $\frac{1}{2}$ or $-\frac{1}{2}$.

32 Using $v = \text{randn}(3, 1)$ in MATLAB, create a random unit vector $u = v/\|v\|$. Using $V = \text{randn}(3, 30)$ create 30 more random unit vectors U_j. What is the average size of the dot products $|u \cdot U_j|$? In calculus, the average $\int_0^\pi |\cos\theta|\,d\theta/\pi = 2/\pi$.